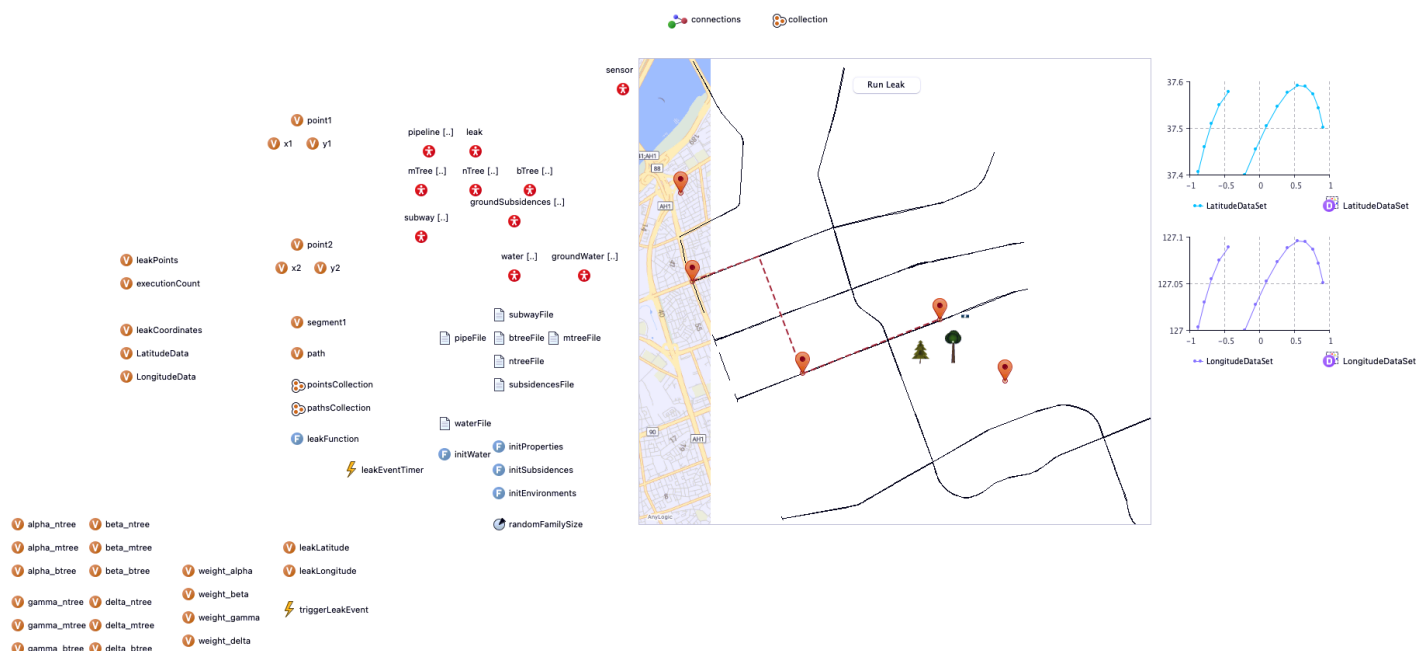


Model: PredictionModelV1

Name	Value
General	
Model time units	minutes
System Dynamics solver	
Differentiation Equations Method	Euler
Algebraic Equations Method	Modified Newton
Mixed Equations Method	RK45+Newton
Absolute accuracy	1.0E-5
Time accuracy	1.0E-5
Relative accuracy	1.0E-5
Fixed time step	0.001
Advanced	
Java package name	predictionmodelv1
File Name	/Users/yschung/Desktop/PredictionModelV1/PredictionModelV1.original.alp

Agent Type: Main

Name	Value
Agent actions	
Startup code	<pre>initEnvironments(); executionCount = 0; // LatitudeDataSet.reset(); // LongitudeDataSet.reset(); // leakPoints.clear(); //</pre>
Agent in flowcharts	
Use in flowcharts as	Agent
Dimensions and movement	
Speed	(10 : MPS)
Rotate animation towards movement	true
Rotate vertically (incline on ramps)	false
Space and network	
Enable steps	false
Advanced Java	
Generic	false
Omniverse	
Usd binding type	None
Advanced	
Flowchart block	false
Logging	true
Auto-create datasets	true
AOC_DATASETS_UPDATE_TIME_PROPERTIES	- Recurring Event Properties
Limit the number of data samples	false



Scale: scale

Name	Value
General	
Unit	meters
Scale	10.0
Type	Defined graphically
Length, pixels	100.0
Show at runtime	false
Lock	false
Public	false
Position and size	
x	0.0
y	-150.0
Rotation	0.0

Parameter: randomFamilySize

Name	Value
General	
Array	false
Default value	uniform_discr(1,7)
Type	int
Show at runtime	true
Show name	true
Value editor	
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Function: initProperties

Name	Value
General	
Return type:	Just action (returns nothing)
Show at runtime	true
Show name	true
Function body	
Body	<pre> /*int maxcolumns = 3; String cols[] = new String[maxcolumns]; int[] skip = {0,0,0}; while(propertiesFile.canReadMore()) { while (propertiesFile.getLineNumber() <= 1) { propertiesFile.skipTokens(1);} System.out.println("Parsing line: " + propertiesFile.getLineNumber()); int id = propertiesFile.readInt(); double lat = propertiesFile.readDouble(); double lon = propertiesFile.readDouble(); Property p = add_properties(id, lat, lon); p.setXY(lat, lon); for (int i=0; i<=randomFamilySize; i++){ Tree pe = add_trees(); p.members.add(pe); pe.jumpTo(lat, lon);} int pos = 0; while (pos < maxcolumns) { if (skip[pos] == 1) { propertiesFile.skipTokens(1);} pos += 1;}}*/ </pre>
Advanced	
Access type	default
System dynamics units	false

Function: leakFunction

Name	Value
General	
Return type:	Just action (returns nothing)
Show at runtime	true
Show name	true
Function body	
Body	<pre> // Define point1 and point2 point1 = new GISPoint(gisMap, true, y1, x1, 0, null, null, 1.0, LINE_STYLE_SOLID, ""); point2 = new GISPoint(gisMap, true, y2, x2, 0, null, null, 1.0, LINE_STYLE_SOLID, ""); // Create a GISMarkupSegmentLine and GISRoute segment1 = new GISMarkupSegmentLine(y1, x1, y2, x2); // Add points and path to collections pointsCollection.add(point1); pointsCollection.add(point2); pathsCollection.add(path); </pre>
Advanced	

Name	Value
Access type	default
System dynamics units	false

Function: initSubsidences

Name	Value
General	
Return type:	Just action (returns nothing)
Show at runtime	true
Show name	true
Function body	
Body	<pre> int maxcolumns = 3; String cols[] = new String[maxcolumns]; int[] skip = {0,0,0}; while(subsidencesFile.canReadMore()) { while (subsidencesFile.getLineNumber() <= 1) { subsidencesFile.skipTokens(1);} System.out.println("Parsing line: " + subsidencesFile.getLineNumber()); int id = subsidencesFile.readInt(); double lat = subsidencesFile.readDouble(); double lon = subsidencesFile.readDouble(); GroundSubsidence p = add_groundSubsidences(id, lat, lon); p.setXY(lat, lon); for (int i=0; i<=randomFamilySize; i++){ GroundSubsidence pe = add_groundSubsidences(); p.members.add(pe); pe.jumpTo(lat, lon);} int pos = 0; while (pos < maxcolumns) { if (skip[pos] == 1) { subsidencesFile.skipTokens(1);} pos += 1;}}</pre>
Advanced	
Access type	default
System dynamics units	false

Function: initEnvironments

Name	Value
General	
Return type:	Just action (returns nothing)
Show at runtime	true
Show name	true
Function body	
Body	<pre> // // Subsidences while (subsidencesFile.canReadMore()) { if (subsidencesFile.getLineNumber() <= 1) { subsidencesFile.skipTokens(3); // () continue; // } try { // int id = subsidencesFile.readInt(); double lat = subsidencesFile.readDouble(); double lon = subsidencesFile.readDouble();</pre>

Name	Value
	<pre> // GroundSubsidence GroundSubsidence gs = add_groundSubsidences(); gs.setXY(lat, lon); // } catch (Exception e) { System.err.println("Error parsing line: " + subsidencesFile.getLineNumber()); continue; } } // btree while (btreeFile.canReadMore()) { if (btreeFile.getLineNumber() <= 1) { btreeFile.skipTokens(3); // (Id, Lat, Lon) . continue; // } try { // String id = btreeFile.readString(); double lon = btreeFile.readDouble(); double lat = btreeFile.readDouble(); // Tree bTree tree = add_bTree(); tree.setXY(lat, lon); // } catch (Exception e) { System.err.println("Error parsing line: " + btreeFile.getLineNumber()); continue; } } // ntree while (ntreeFile.canReadMore()) { if (ntreeFile.getLineNumber() <= 1) { ntreeFile.skipTokens(3); // (Id, Lat, Lon) . continue; // } try { // String id = ntreeFile.readString(); double lon = ntreeFile.readDouble(); double lat = ntreeFile.readDouble(); // Tree nTree tree = add_nTree(); tree.setXY(lat, lon); // } catch (Exception e) { System.err.println("Error parsing line: " + ntreeFile.getLineNumber()); continue; } } // mtree while (mtreeFile.canReadMore()) { if (mtreeFile.getLineNumber() <= 1) { mtreeFile.skipTokens(3); // (Id, Lat, Lon) . continue; // } try { // String id = mtreeFile.readString(); double lon = mtreeFile.readDouble(); double lat = mtreeFile.readDouble(); // Tree </pre>

Name	Value
	<pre> mTree tree = add_mTree(); tree.setXY(lat, lon); // } catch (Exception e) { System.err.println("Error parsing line: " + mtreeFile. getLineNumber()); continue; } } // pipes while (pipeFile.canReadMore()) { if (pipeFile.getLineNumber() <= 1) { pipeFile.skipTokens(4); // () continue; // } try { // int id = pipeFile.readInt(); double lat = pipeFile.readDouble(); double lon = pipeFile.readDouble(); int year = pipeFile.readInt(); // subwayline Pipeline pl = add_pipeline(); pl.id = id; pl.lat = lat; pl.lon = lon; pl.year = year; pl.setXY(lat, lon); // (ID) } catch (Exception e) { System.err.println("Error parsing line: " + pipeFile.getLineNumber()); continue; } } // Subway () List<List<double[]>> subwayLines = new ArrayList<>(); // while (subwayFile.canReadMore()) { if (subwayFile.getLineNumber() <= 1) { subwayFile.skipTokens(4); // continue; } try { // int id = subwayFile.readInt(); // ID double lat = subwayFile.readDouble(); // double lon = subwayFile.readDouble(); // double length = subwayFile.readDouble(); // // subwayline Subway sub = add_subway(); sub.id = id; sub.lat = lat; sub.lon = lon; sub.length = length; sub.setXY(lat, lon); // // (ID) while (subwayLines.size() <= id) { subwayLines.add(new ArrayList<>()); } subwayLines.get(id).add(new double[]{lat, lon, length}); } catch (Exception e) { System.err.println("Error parsing subway data: " + subwayFile. getLineNumber()); } } </pre>

Name	Value
Advanced	
Access type	default
System dynamics units	false

Function: initWater

Name	Value
General	
Return type:	Just action (returns nothing)
Show at runtime	true
Show name	true
Function body	
Body	<pre> /*int maxcolumns = 4; String cols[] = new String[maxcolumns]; int[] skip = {0,0,0,0}; while(waterFile.canReadMore()) { while (waterFile.getLineNumber() <= 1) { waterFile.skipTokens(1);} System.out.println("Parsing line: " + waterFile.getLineNumber()); int id = waterFile.readInt(); double Elev = waterFile.readDouble(); double lat = waterFile.readDouble(); double lon = waterFile.readDouble(); Water p = add_water(id, Elev, lat, lon); p.setXYZ(lat, lon, Elev); for (int i=0; i<=randomFamilySize; i++){ GroundWater pe = add_groundWater(); p.members.add(pe); pe.jumpTo(lat, lon, Elev);} int pos = 0; while (pos < maxcolumns) { if (skip[pos] == 1) { waterFile.skipTokens(1);} pos += 1;}} */ </pre>
Advanced	
Access type	default
System dynamics units	false

Event: triggerLeakEvent

Name	Value
General	
Logging	false
EVENT_TIMEOUT_PROPERTIES	- Recurring Event Properties
Mode	Cyclic
Trigger type	Timeout
Show at runtime	true
Show name	true
Action	
Action	<pre> // Subsidences List<double[]> subsidencesList = new ArrayList<>(); // Subsidences while (subsidencesFile.canReadMore()) { if (subsidencesFile.getLineNumber() <= 1) { subsidencesFile.skipTokens(3); // </pre>

Name	Value
	<pre> continue; } try { double lat = subsidencesFile.readDouble(); double lon = subsidencesFile.readDouble(); subsidencesList.add(new double[]{lat, lon}); // Subsidences } catch (Exception e) { System.err.println("Error reading subsidences: " + e.getMessage()); } } /*System.out.println("Subsidences loaded: " + subsidencesList.size());*/ // TriggerLeakEvent if (executionCount < 500) { // 1000 int currentYear = 2024 + (int)time(); // Pipeline leakPipe = null; // double highestProbability = 0; // // for (Pipeline pipe : pipeline) { int pipeAge = currentYear - pipe.year; double baseRuptureProbability = Math.min(0.8, pipeAge * 0.02); double treeRisk = 0; // Ginkgo (ntree) treeRisk += nTree.size() * (alpha_ntree * weight_alpha + beta_ntree * weight_beta + gamma_ntree * weight_gamma + delta_ntree * weight_delta); // Zelkova (btree) treeRisk += bTree.size() * (alpha_btree * weight_alpha + beta_btree * weight_beta + gamma_btree * weight_gamma + delta_btree * weight_delta); // Metasequoia (mtree) treeRisk += mTree.size() * (alpha_mtree * weight_alpha + beta_mtree * weight_beta + gamma_mtree * weight_gamma + delta_mtree * weight_delta); double totalRuptureProbability = baseRuptureProbability + treeRisk + Math.random() * 0.1; if (totalRuptureProbability > highestProbability) { highestProbability = totalRuptureProbability; leakPipe = pipe; } } if (leakPipe != null) { double leakLatitude = leakPipe.lat; // Leak double leakLongitude = leakPipe.lon; // Leak // Leak leak.jumpTo(leakLatitude, leakLongitude); // DataSet LatitudeDataSet.add(time(), leakLatitude); // LongitudeDataSet.add(time(), leakLongitude); // // Subsidences double matchThreshold = 0.01; // 1km int matches = 0; for (double[] subsidences : subsidencesList) { double subLat = subsidences[0]; double subLon = subsidences[1]; </pre>

Name	Value
	<pre> // double latDiff = Math.abs(leakLatitude - subLat); double lonDiff = Math.abs(leakLongitude - subLon); if (latDiff <= matchThreshold && lonDiff <= matchThreshold) { matches++; } } // System.out.println("Execution #" + executionCount + ": Leak occurred at (" + leakLatitude + ", " + leakLongitude); /*System.out.println("Matching subsidences count: " + matches);*/ // executionCount++; } else { System.out.println("No pipeline eligible for leak event."); } } else { System.out.println("Maximum number of leak events reached."); stopSimulation(); // } </pre>

Event: leakEventTimer

Name	Value
General	
Logging	false
EVENT_TIMEOUT_PROPERTIES	- Recurring Event Properties
Mode	Cyclic
Trigger type	Timeout
Show at runtime	true
Show name	true
Action	
Action	<pre> /*if (executionCount < 30) { // 30 int currentYear = 2024 + (int)time(); // Pipeline leakPipe = null; // double highestProbability = 0; // // for (Pipeline pipe : pipeline) { int pipeAge = currentYear - pipe.year; // double ruptureProbability = Math.random() * Math.min(0.8, pipeAge * 0.02); // // if (ruptureProbability > highestProbability) { highestProbability = ruptureProbability; leakPipe = pipe; // Leak } } // LeakEvent if (leakPipe != null) { double leakLatitude = leakPipe.lat; // double leakLongitude = leakPipe.lon; // // Leak leak.jumpTo(leakLongitude, leakLatitude); // GISPoint point = gisMap.addPoint(leakLatitude, leakLongitude); // </pre>

Name	Value
	<pre> GISPoint leakPoints.add(point); // // Data Set LatitudeDataSet.add(time(), leakLatitude); // LongitudeDataSet.add(time(), leakLongitude); // // executionCount++; // System.out.println(currentYear + ": Execution #" + executionCount + ": Leak occurred at Pipeline ID=" + leakPipe.id + " (" + leakLatitude + ", " + leakLongitude + ")"); } else { System.out.println("No pipeline eligible for LeakEvent."); } } else { System.out.println("Maximum number of leak events (30) reached."); stopSimulation(); // }*/ </pre>

Variable: point1

Name	Value
General	
Type	GISPoint
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: point2

Name	Value
General	
Type	GISPoint
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: x1

Name	Value
General	
Initial value	gisPoint.getLongitude()
Type	double

Name	Value
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: y1

Name	Value
General	
Initial value	gisPoint.getLatitude()
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: x2

Name	Value
General	
Initial value	gisPoint1.getLongitude()
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: y2

Name	Value
General	
Initial value	gisPoint1.getLatitude()
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: segment1

Name	Value
General	
Type	GISMarkupSegmentLine
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: path

Name	Value
General	
Type	GISRoute
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: leakLatitude

Name	Value
General	
Initial value	0.0
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: leakLongitude

Name	Value
General	
Initial value	0.0
Type	double
Show at runtime	true
Show name	true
Advanced	

Name	Value
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: leakCoordinates

Name	Value
General	
Initial value	new ArrayList<>()
Type	List<Pair<Double, Double>>
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: LatitudeData

Name	Value
General	
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: LongitudeData

Name	Value
General	
Initial value	0
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: executionCount

Name	Value
------	-------

Name	Value
General	
Initial value	0
Type	int
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: leakPoints

Name	Value
General	
Initial value	new ArrayList<>()
Type	List<GISPoint>
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: alpha_ntree

Name	Value
General	
Initial value	3
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: alpha_mtree

Name	Value
General	
Initial value	5
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public

Name	Value
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: alpha_btree

Name	Value
General	
Initial value	4
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: beta_ntree

Name	Value
General	
Initial value	1
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: beta_btree

Name	Value
General	
Initial value	3
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: beta_mtree

Name	Value
------	-------

Name	Value
General	
Initial value	3
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: gamma_ntree

Name	Value
General	
Initial value	1
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: gamma_mtree

Name	Value
General	
Initial value	5
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: gamma_btree

Name	Value
General	
Initial value	5
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public

Name	Value
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: delta_ntree

Name	Value
General	
Initial value	5
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: delta_mtree

Name	Value
General	
Initial value	5
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: delta_btree

Name	Value
General	
Initial value	5
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: weight_alpha

Name	Value
------	-------

Name	Value
General	
Initial value	0.3
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: weight_beta

Name	Value
General	
Initial value	0.2
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: weight_gamma

Name	Value
General	
Initial value	0.3
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: weight_delta

Name	Value
General	
Initial value	0.2
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public

Name	Value
Constant	false
Save in snapshot	true
System dynamics units	false

Collection: pointsCollection

Name	Value
General	
Initial contents	{}
Initial contents	{}
Element class	GISPoint
Collection class	ArrayList
Show at runtime	true
Show name	true
Advanced	
Access type	public
Save in snapshot	true

Collection: pathsCollection

Name	Value
General	
Initial contents	{}
Initial contents	{}
Element class	GISRoute
Collection class	ArrayList
Show at runtime	true
Show name	true
Advanced	
Access type	public
Save in snapshot	true

Collection: collection

Name	Value
General	
Initial contents	{LatitudeDataSet, LongitudeDataSet}
Initial contents	{LatitudeDataSet, LongitudeDataSet}
Element class	DataSet
Collection class	ArrayList
Show at runtime	true
Show name	true
Advanced	
Access type	public
Save in snapshot	true

Plot: plot

Name	Value
General	
Lock	false
Public	true
Data update	
Analysis auto update	true
ANALYSIS_UPDATE_TIME_PROPERTIES	- Recurring Event Properties
Dataset Samples To Keep	500
Scale	
Horizontal scale	Auto
Vertical scale	Fixed
Chart Vertical Scale: From	37.4
Chart Vertical Scale: To	37.6
Appearance	
Labels horizontal position	DEFAULT
Labels vertical position	DEFAULT
Labels Text Color	darkGray
Chart Area Grid Color	darkGray
Draw line	true
Interpolation	Linear
Position and size	
x	660.0
Width	260.0
y	0.0
Height	210.0
Legend	
Show legend	true
Legend size	30.0
Legend text color	black
Chart area	
Chart Area: X Offset	50.0
Chart Area: Width	180.0
Chart Area: Y Offset	30.0
Chart Area: Height	120.0
Chart Area: Background Color	white
Chart area border color	black
Omniverse	
Usd binding type	None
Advanced	
Show name	false
Logging	true

Plot Items:

Title	Type	Dataset		Point Style	Color	Line	Width	Interpolation
		X Axis Value	Y Axis Value					
LatitudeDataSet	dataset	LatitudeDataSet		CIRCLE	deepSkyBlue	true	1.0	LINEAR

Plot: plot1

Name	Value
------	-------

Name	Value
General	
Lock	false
Public	true
Data update	
Analysis auto update	true
ANALYSIS_UPDATE_TIME_PROPERTIES	- Recurring Event Properties
Dataset Samples To Keep	500
Scale	
Horizontal scale	Auto
Vertical scale	Fixed
Chart Vertical Scale: From	127
Chart Vertical Scale: To	127.1
Appearance	
Labels horizontal position	DEFAULT
Labels vertical position	DEFAULT
Labels Text Color	darkGray
Chart Area Grid Color	darkGray
Draw line	true
Interpolation	Linear
Position and size	
x	660.0
Width	260.0
y	200.0
Height	210.0
Legend	
Show legend	true
Legend size	30.0
Legend text color	black
Chart area	
Chart Area: X Offset	50.0
Chart Area: Width	180.0
Chart Area: Y Offset	30.0
Chart Area: Height	120.0
Chart Area: Background Color	white
Chart area border color	black
Omniverse	
Usd binding type	None
Advanced	
Show name	false
Logging	true

Plot Items:

Title	Type	Dataset		Point Style	Color	Line	Width	Interpolation
		X Axis Value	Y Axis Value					
LongitudeDataSet	dataset	LongitudeDataSet		CIRCLE	lightSlateBlue	true	1.0	LINEAR

Data Set: LatitudeDataSet

Name	Value
General	
Dataset Samples To Keep	500
Axis Data Vertical Y Axis	leakLatitude
Axis Data Freeze X Axis	true
Show at runtime	true
Show name	true
Data update	
Analysis auto update	true
ANALYSIS_UPDATE_TIME_PROPERTIES	- Recurring Event Properties
Logging	true

Data Set: LongitudeDataSet

Name	Value
General	
Dataset Samples To Keep	500
Axis Data Vertical Y Axis	leakLongitude
Axis Data Freeze X Axis	true
Show at runtime	true
Show name	true
Data update	
Analysis auto update	true
ANALYSIS_UPDATE_TIME_PROPERTIES	- Recurring Event Properties
Logging	true

Agent Presentation: leak_presentation

Name	Value
General	
Show at runtime	true
Public	true
Position and size	
Real size when map scale is under	1000
Fixed size when map scale is under	1000000000
Omniverse	
Usd binding type	None
Advanced	
Show in	2D and 3D
Show name	false

Agent Presentation: sensor_presentation

Name	Value
General	
Show at runtime	true
Public	true
Position and size	
Real size when map scale is under	1000
Fixed size when map scale is under	1000000000

Name	Value
Omniverse	
Usd binding type	None
Advanced	
Show in	2D and 3D
Show name	false

Agent Presentation: groundSubsidences_presentation

Name	Value
General	
Show at runtime	true
Public	true
Position and size	
Real size when map scale is under	1000
Fixed size when map scale is under	1000000000
Omniverse	
Usd binding type	None
Advanced	
Show in	2D and 3D
Show name	false

Agent Presentation: water_presentation

Name	Value
General	
Show at runtime	true
Public	true
Position and size	
Real size when map scale is under	1000
Fixed size when map scale is under	1000000000
Omniverse	
Usd binding type	None
Advanced	
Show in	2D and 3D
Show name	false

Agent Presentation: groundWater_presentation

Name	Value
General	
Show at runtime	true
Public	true
Position and size	
Real size when map scale is under	1000
Fixed size when map scale is under	1000000000
Omniverse	
Usd binding type	None
Advanced	
Show in	2D and 3D

Name	Value
Show name	false

Agent Presentation: pipeline_presentation

Name	Value
General	
Show at runtime	true
Public	true
Position and size	
Real size when map scale is under	1000
Fixed size when map scale is under	1000000000
Omniverse	
Usd binding type	None
Advanced	
Show in	2D and 3D
Show name	false

Agent Presentation: nTree_presentation

Name	Value
General	
Show at runtime	true
Public	true
Position and size	
Real size when map scale is under	1000
Fixed size when map scale is under	1000000000
Omniverse	
Usd binding type	None
Advanced	
Show in	2D and 3D
Show name	false

Agent Presentation: bTree_presentation

Name	Value
General	
Show at runtime	true
Public	true
Position and size	
Real size when map scale is under	1000
Fixed size when map scale is under	1000000000
Omniverse	
Usd binding type	None
Advanced	
Show in	2D and 3D
Show name	false

GIS Map: gisMap

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Routing	
Routes are	Requested from OSM server
Routing method	Shortest
Road type	Foot
If route not found	Create straight route
Center and scale	
Latitude	37.51
Longitude	127.05
Scale	40000
Appearance	
Border color	silver
Fill color	white
Position and size	
x	0.0
Width	660.0
y	0.0
Height	600.0
Omniverse	
Usd binding type	None
Advanced	
Show tiles	true
Use custom tile URL	false
Use custom route provider	false
Routes and regions generalization uses Current map scale	true

Shape Files:

Shape File	DBF	Name Column	Default Line Color	Default Fill Color
_.shp	_.dbf	0	black	white

Button: button

Name	Value
General	
Enabled	true
Label text	Run Leak
Lock	false
Public	true
Action	
Action	<pre> /*if (executionCount < 500) { // 30 int currentYear = 2024 + (int)time(); // Pipeline leakPipe = null; // double highestProbability = 0; // // for (Pipeline pipe : pipeline) { int pipeAge = currentYear - pipe.year; // </pre>

Name	Value
	<pre> double ruptureProbability = Math.random() * Math.min(0.8, pipeAge * 0.02); // // if (ruptureProbability > highestProbability) { highestProbability = ruptureProbability; leakPipe = pipe; // Leak } } // LeakEvent if (leakPipe != null) { double leakLatitude = leakPipe.lat; // double leakLongitude = leakPipe.lon; // // leak.jumpTo(leakLatitude,leakLongitude); // Longitude X // Latitude Y // // LatitudeDataSet.add(time(), leakLatitude); // LongitudeDataSet.add(time(), leakLongitude); // // executionCount++; // System.out.println(currentYear + ": Execution #" + executionCount + ": Leak occurred at Pipeline ID=" + leakPipe.id + " (" + leakLatitude + ", " + leakLongitude + ")"); } else { System.out.println("No pipeline eligible for LeakEvent."); } } else { System.out.println("Maximum number of leak events (500) reached."); stopSimulation(); // }*/ </pre>
Appearance	
Text color	black
Position and size	
x	270.0
Width	100.0
y	20.0
Height	30.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false
Embedded icon	false

Text File: btreeFile

Name	Value
General	
Text file separators	[C@2acc6c58
File mode	Read
Text file name	ResourceReference: predictionmodelv1.btree.csv (Resolved: true)
Text File Type	File

Name	Value
Show at runtime	true
Show name	true

Text File: subsidencesFile

Name	Value
General	
Text file separators	[C@363679a
File mode	Read
Text file name	ResourceReference: predictionmodelv1.groundsubsidences.csv (Resolved: true)
Text File Type	File
Show at runtime	true
Show name	true

Text File: waterFile

Name	Value
General	
Text file separators	[C@6b1744b0
File mode	Read
Text File Type	File
Show at runtime	true
Show name	true

Text File: pipeFile

Name	Value
General	
Text file separators	[C@241f9269
File mode	Read
Text file name	ResourceReference: predictionmodelv1.pipe5.csv (Resolved: true)
Text File Type	File
Show at runtime	false
Show name	true

Text File: ntreeFile

Name	Value
General	
Text file separators	[C@6d78841e
File mode	Read
Text file name	ResourceReference: predictionmodelv1.ntree.csv (Resolved: true)
Text File Type	File
Show at runtime	true
Show name	true

Text File: mtreeFile

Name	Value
General	
Text file separators	[C@7c08cb77
File mode	Read
Text file name	ResourceReference: predictionmodelv1.mtree.csv (Resolved: true)
Text File Type	File
Show at runtime	true
Show name	true

Text File: subwayFile

Name	Value
General	
Text file separators	[C@55f9f89c
File mode	Read
Text file name	ResourceReference: predictionmodelv1.subway.csv (Resolved: true)
Text File Type	File
Show at runtime	true
Show name	true

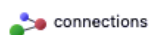
Link to agents: connections

Name	Value
General	
Show at runtime	true
Show name	true
Communication	
Message type	Object
Animation	
Draw line	false

Agent Type: Leak

Name	Value
Agent in flowcharts	
Use in flowcharts as	Agent
Dimensions and movement	
Speed	(10 : MPS)
Rotate animation towards movement	true
Rotate vertically (incline on ramps)	false
Space and network	
Space Type	Continuous
Layout Type	User-defined
Network type	User-defined
Enable steps	false
Advanced Java	
Generic	false
Omniverse	
Usd binding type	None
Advanced	

Name	Value
Flowchart block	false
Logging	true
Auto-create datasets	true
AOC_DATASETS_UPDATE_TIME_PROPERTIES	- Recurring Event Properties
Limit the number of data samples	false



Scale: scale

Name	Value
General	
Unit	meters
Scale	10.0
Type	Defined graphically
Length, pixels	100.0
Show at runtime	false
Lock	false
Public	false
Position and size	
x	0.0
y	-150.0
Rotation	0.0

3D Object: pipe_junction

Name	Value
General	
Scale	1.5
Auto scale	true
File Name	ResourceReference: predictionmodelv1.3d/pipe_junction.dae (Resolved: true)
Show at runtime	true
Lock	false
Public	true
Position and size	
x	0.0
y	0.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None

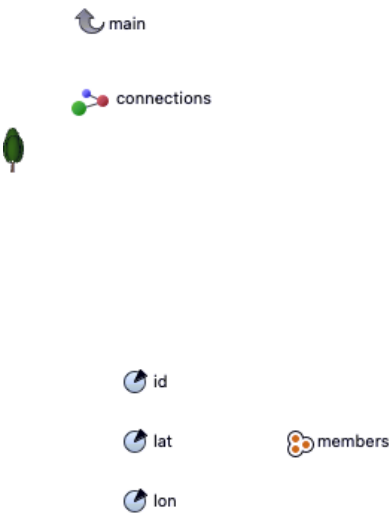
Name	Value
Usd ignore generated object	false
Advanced	
Show in	2D and 3D
Show name	false
Internal lights	Off

Link to agents: connections

Name	Value
General	
Show at runtime	true
Show name	true
Communication	
Message type	Object
Animation	
Draw line	false

Agent Type: mTree

Name	Value
Agent in flowcharts	
Use in flowcharts as	Agent
Dimensions and movement	
Speed	(10 : MPS)
Rotate animation towards movement	true
Rotate vertically (incline on ramps)	false
Space and network	
Space Type	Continuous
Layout Type	User-defined
Network type	User-defined
Enable steps	false
Advanced Java	
Generic	false
Omniverse	
Usd binding type	None
Advanced	
Flowchart block	false
Logging	true
Auto-create datasets	true
AOC_DATASETS_UPDATE_TIME_PROPERTIES	- Recurring Event Properties
Limit the number of data samples	false



Scale: scale

Name	Value
General	
Unit	meters
Scale	10.0
Type	Defined graphically
Length, pixels	100.0
Show at runtime	false
Lock	false
Public	false
Position and size	
x	0.0
y	-150.0
Rotation	0.0

Parameter: id

Name	Value
General	
Array	false
Type	String
Show at runtime	true
Show name	true
Value editor	
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Parameter: lat

Name	Value
General	
Array	false

Name	Value
Type	double
Show at runtime	true
Show name	true
Value editor	
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Parameter: lon

Name	Value
General	
Array	false
Type	double
Show at runtime	true
Show name	true
Value editor	
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Collection: members

Name	Value
General	
Initial contents	{}
Initial contents	{}
Element class	mTree
Collection class	ArrayList
Show at runtime	true
Show name	true
Advanced	
Access type	public
Save in snapshot	true

3D Object: tree_3

Name	Value
General	
Scale	0.35
Auto scale	true
File Name	ResourceReference: predictionmodelv1.3d/tree_3.dae (Resolved: true)
Show at runtime	true
Lock	false
Public	true

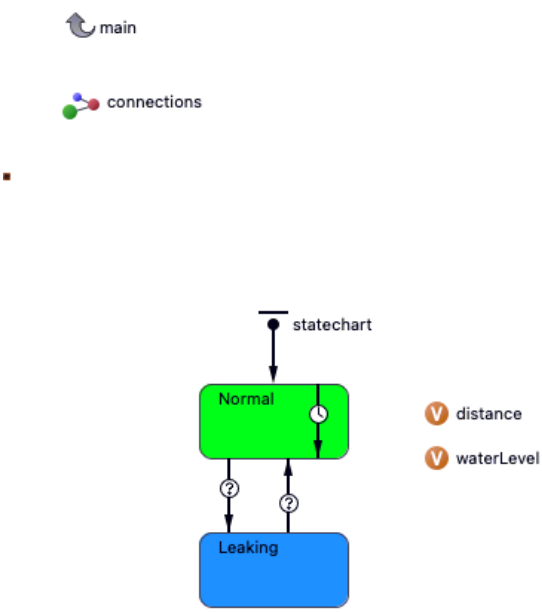
Name	Value
Position and size	
x	0.0
y	0.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Usd ignore generated object	false
Advanced	
Show in	2D and 3D
Show name	false
Internal lights	Off

Link to agents: connections

Name	Value
General	
Show at runtime	true
Show name	true
Communication	
Message type	Object
Animation	
Draw line	false

Agent Type: Sensor

Name	Value
Agent in flowcharts	
Use in flowcharts as	Agent
Dimensions and movement	
Speed	(10 : MPS)
Rotate animation towards movement	true
Rotate vertically (incline on ramps)	false
Space and network	
Space Type	Continuous
Layout Type	User-defined
Network type	User-defined
Enable steps	false
Advanced Java	
Generic	false
Omniverse	
Usd binding type	None
Advanced	
Flowchart block	false
Logging	true
Auto-create datasets	true
AOC_DATASETS_UPDATE_TIME_PROPERTIES	- Recurring Event Properties
Limit the number of data samples	false



Scale: scale

Name	Value
General	
Unit	meters
Scale	10.0
Type	Defined graphically
Length, pixels	100.0
Show at runtime	false
Lock	false
Public	false
Position and size	
x	0.0
y	-150.0
Rotation	0.0

Variable: distance

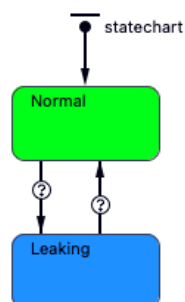
Name	Value
General	
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: waterLevel

Name	Value
General	
Initial value	1
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Statechart Entry Point: statechart

Name	Value
General	
Logging	true
Show at runtime	true
Show name	true



Transition: sensing

Name	Value
General	
Action	//calculates distance between the leak and the sensor distance = distanceTo(main.leak); // sets the condition. If distance to leak is less than 10 it sets the water level from 1 (no leak) to leak level (0) if (distance < 10) waterLevel=0;
Timeout	(1 : SECOND)
Trigger type	Timeout
Show name	false

Transition: leaking

Name	Value
General	
Action	/*//this increases the size of sensor cone.setScale(3); //The following lines send a message to all people. for (Tree per:main.trees){

Name	Value
	<pre>per.receive("leakWarning"); }*/</pre>
Condition	<code>waterLevel==0;</code>
Trigger type	Condition
Show name	false

Transition: transition

Name	Value
General	
Action	<pre>/*//this line reduces the size of the sensor symbole to 1 cone.setScale(1); //These lines inform people that leak situation is normal. for (Tree p:main.trees){ p.receive("normal"); }*/</pre>
Condition	<code>waterLevel==1;</code>
Trigger type	Condition
Show name	false

State: Normal

Name	Value
General	
Fill color	lime
Show name	true

State: Leaking

Name	Value
General	
Fill color	dodgerBlue
Show name	true

3D Object: cone

Name	Value
General	
Scale	1.0
Auto scale	true
File Name	ResourceReference: predictionmodelv1.3d/traffic_cone.dae (Resolved: true)
Show at runtime	true
Lock	false
Public	true
Position and size	
x	0.0
y	0.0
z	0.0
Rotation	0.0

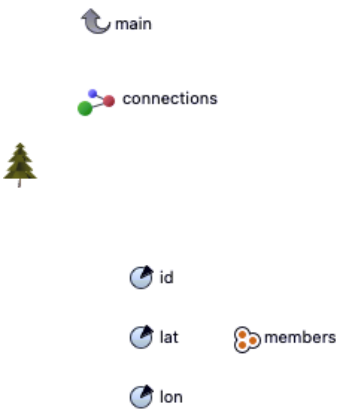
Name	Value
Omniverse	
Usd binding type	None
Usd ignore generated object	false
Advanced	
Show in	2D and 3D
Show name	false
Internal lights	Off

Link to agents: connections

Name	Value
General	
Show at runtime	true
Show name	true
Communication	
Message type	Object
Animation	
Draw line	false

Agent Type: bTree

Name	Value
Agent in flowcharts	
Use in flowcharts as	Agent
Dimensions and movement	
Speed	(10 : MPS)
Rotate animation towards movement	true
Rotate vertically (incline on ramps)	false
Space and network	
Space Type	Continuous
Layout Type	User-defined
Network type	User-defined
Enable steps	false
Advanced Java	
Generic	false
Omniverse	
Usd binding type	None
Advanced	
Flowchart block	false
Logging	true
Auto-create datasets	true
AOC_DATASETS_UPDATE_TIME_PROPERTIES	- Recurring Event Properties
Limit the number of data samples	false



Scale: scale

Name	Value
General	
Unit	meters
Scale	10.0
Type	Defined graphically
Length, pixels	100.0
Show at runtime	false
Lock	false
Public	false
Position and size	
x	0.0
y	-150.0
Rotation	0.0

Parameter: id

Name	Value
General	
Array	false
Type	String
Show at runtime	true
Show name	true
Value editor	
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Parameter: lat

Name	Value
General	
Array	false
Type	double
Show at runtime	true
Show name	true

Name	Value
Value editor	
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Parameter: lon

Name	Value
General	
Array	false
Type	double
Show at runtime	true
Show name	true
Value editor	
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Collection: members

Name	Value
General	
Initial contents	{}
Initial contents	{}
Element class	bTree
Collection class	ArrayList
Show at runtime	true
Show name	true
Advanced	
Access type	public
Save in snapshot	true

3D Object: tree_1

Name	Value
General	
Scale	0.35
Auto scale	true
File Name	ResourceReference: predictionmodelv1.3d/tree_1.dae (Resolved: true)
Show at runtime	true
Lock	false
Public	true
Position and size	
x	0.0
y	10.0

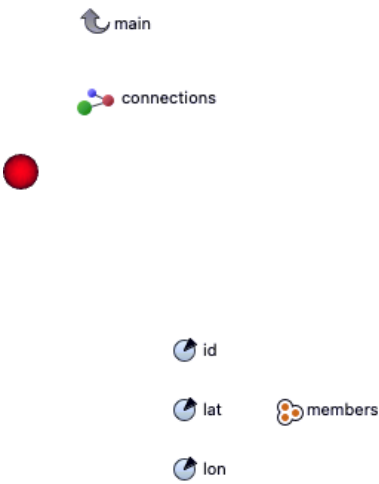
Name	Value
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Usd ignore generated object	false
Advanced	
Show in	2D and 3D
Show name	false
Internal lights	Off

Link to agents: connections

Name	Value
General	
Show at runtime	true
Show name	true
Communication	
Message type	Object
Animation	
Draw line	false

Agent Type: GroundSubsidence

Name	Value
Agent in flowcharts	
Use in flowcharts as	Agent
Dimensions and movement	
Speed	(10 : MPS)
Rotate animation towards movement	true
Rotate vertically (incline on ramps)	false
Space and network	
Space Type	Continuous
Layout Type	User-defined
Network type	User-defined
Enable steps	false
Advanced Java	
Generic	false
Omniverse	
Usd binding type	None
Advanced	
Flowchart block	false
Logging	true
Auto-create datasets	true
AOC_DATASETS_UPDATE_TIME_PROPERTIES	- Recurring Event Properties
Limit the number of data samples	false



Scale: scale

Name	Value
General	
Unit	meters
Scale	10.0
Type	Defined graphically
Length, pixels	100.0
Show at runtime	false
Lock	false
Public	false
Position and size	
x	0.0
y	-150.0
Rotation	0.0

Parameter: id

Name	Value
General	
Array	false
Type	int
Show at runtime	true
Show name	true
Value editor	
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Parameter: lat

Name	Value
General	
Array	false
Type	double

Name	Value
Show at runtime	true
Show name	true
Value editor	
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Parameter: lon

Name	Value
General	
Array	false
Type	double
Show at runtime	true
Show name	true
Value editor	
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Collection: members

Name	Value
General	
Initial contents	{}
Initial contents	{}
Element class	GroundSubsidence
Collection class	ArrayList
Show at runtime	true
Show name	true
Advanced	
Access type	public
Save in snapshot	true

3D Object: sphere

Name	Value
General	
Scale	2.0
Auto scale	true
File Name	ResourceReference: predictionmodelv1.3d/sphere.dae (Resolved: true)
Show at runtime	true
Lock	false
Public	true
Position and size	

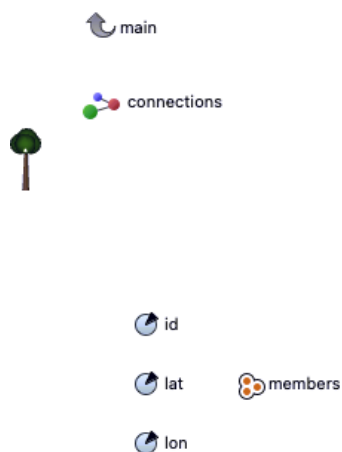
Name	Value
x	0.0
y	0.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Usd ignore generated object	false
Advanced	
Show in	2D and 3D
Show name	false
Internal lights	Off

Link to agents: connections

Name	Value
General	
Show at runtime	true
Show name	true
Communication	
Message type	Object
Animation	
Draw line	false

Agent Type: nTree

Name	Value
Agent in flowcharts	
Use in flowcharts as	Agent
Dimensions and movement	
Speed	(10 : MPS)
Rotate animation towards movement	true
Rotate vertically (incline on ramps)	false
Space and network	
Space Type	Continuous
Layout Type	User-defined
Network type	User-defined
Enable steps	false
Advanced Java	
Generic	false
Omniverse	
Usd binding type	None
Advanced	
Flowchart block	false
Logging	true
Auto-create datasets	true
AOC_DATASETS_UPDATE_TIME_PROPERTIES	- Recurring Event Properties
Limit the number of data samples	false



Scale: scale

Name	Value
General	
Unit	meters
Scale	10.0
Type	Defined graphically
Length, pixels	100.0
Show at runtime	false
Lock	false
Public	false
Position and size	
x	0.0
y	-150.0
Rotation	0.0

Parameter: id

Name	Value
General	
Array	false
Type	String
Show at runtime	true
Show name	true
Value editor	
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Parameter: lat

Name	Value
General	
Array	false
Type	double

Name	Value
Show at runtime	true
Show name	true
Value editor	
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Parameter: lon

Name	Value
General	
Array	false
Type	double
Show at runtime	true
Show name	true
Value editor	
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Collection: members

Name	Value
General	
Initial contents	{}
Initial contents	{}
Element class	nTree
Collection class	ArrayList
Show at runtime	true
Show name	true
Advanced	
Access type	public
Save in snapshot	true

3D Object: tree_2

Name	Value
General	
Scale	0.5
Auto scale	true
File Name	ResourceReference: predictionmodelv1.3d/tree_2.dae (Resolved: true)
Show at runtime	true
Lock	false
Public	true
Position and size	

Name	Value
x	0.0
y	10.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Usd ignore generated object	false
Advanced	
Show in	2D and 3D
Show name	false
Internal lights	Off

Link to agents: connections

Name	Value
General	
Show at runtime	true
Show name	true
Communication	
Message type	Object
Animation	
Draw line	false

Agent Type: Water

Name	Value
Agent in flowcharts	
Use in flowcharts as	Agent
Dimensions and movement	
Speed	(10 : MPS)
Rotate animation towards movement	true
Rotate vertically (incline on ramps)	false
Space and network	
Space Type	Continuous
Layout Type	User-defined
Network type	User-defined
Enable steps	false
Advanced Java	
Generic	false
Omniverse	
Usd binding type	None
Advanced	
Flowchart block	false
Logging	true
Auto-create datasets	true
AOC_DATASETS_UPDATE_TIME_PROPERTIES	- Recurring Event Properties
Limit the number of data samples	false



Scale: scale

Name	Value
General	
Unit	meters
Scale	10.0
Type	Defined graphically
Length, pixels	100.0
Show at runtime	false
Lock	false
Public	false
Position and size	
x	0.0
y	-150.0
Rotation	0.0

Parameter: id

Name	Value
General	
Array	false
Type	double
Show at runtime	true
Show name	true
Value editor	
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Parameter: elev

Name	Value
General	
Array	false
Type	double
Show at runtime	true
Show name	true
Value editor	
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Parameter: lat

Name	Value
General	
Array	false
Type	double
Show at runtime	true
Show name	true
Value editor	
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Parameter: lon

Name	Value
General	
Array	false
Type	double
Show at runtime	true
Show name	true
Value editor	
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Collection: members

Name	Value
General	
Initial contents	{}
Initial contents	{}
Element class	Water
Collection class	ArrayList
Show at runtime	true

Name	Value
Show name	true
Advanced	
Access type	public
Save in snapshot	true

3D Object: sphere

Name	Value
General	
Scale	0.5
Auto scale	true
File Name	ResourceReference: predictionmodelv1.3d/sphere.dae (Resolved: true)
Show at runtime	true
Lock	false
Public	true
Position and size	
x	0.0
y	0.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Usd ignore generated object	false
Advanced	
Show in	2D and 3D
Show name	false
Internal lights	Off

Link to agents: connections

Name	Value
General	
Show at runtime	true
Show name	true
Communication	
Message type	Object
Animation	
Draw line	false

Agent Type: Pipeline

Name	Value
Agent in flowcharts	
Use in flowcharts as	Agent
Dimensions and movement	
Speed	(10 : MPS)
Rotate animation towards movement	true
Rotate vertically (incline on ramps)	false
Space and network	

Name	Value
Space Type	Continuous
Layout Type	User-defined
Network type	User-defined
Enable steps	false
Advanced Java	
Generic	false
Omniverse	
Usd binding type	None
Advanced	
Flowchart block	false
Logging	true
Auto-create datasets	true
AOC_DATASETS_UPDATE_TIME_PROPERTIES	- Recurring Event Properties
Limit the number of data samples	false



main



connections



id



lat



members



lon



year

Scale: scale

Name	Value
General	
Unit	meters
Scale	10.0
Type	Defined graphically
Length, pixels	100.0
Show at runtime	false
Lock	false
Public	false
Position and size	
x	0.0
y	-150.0
Rotation	0.0

Parameter: id

Name	Value
General	
Array	false
Type	double
Show at runtime	true
Show name	true
Value editor	
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Parameter: lat

Name	Value
General	
Array	false
Type	double
Show at runtime	true
Show name	true
Value editor	
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Parameter: lon

Name	Value
General	
Array	false
Type	double
Show at runtime	true
Show name	true
Value editor	
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Parameter: year

Name	Value
General	
Array	false
Type	int
Show at runtime	true
Show name	true
Value editor	
Editor control	Text

Name	Value
Advanced	
System dynamics units	false
Save in snapshot	true

Collection: members

Name	Value
General	
Initial contents	{}
Initial contents	{}
Element class	Pipeline
Collection class	ArrayList
Show at runtime	true
Show name	true
Advanced	
Access type	public
Save in snapshot	true

3D Object: pipe_junction

Name	Value
General	
Scale	0.5
Auto scale	true
File Name	ResourceReference: predictionmodelv1.3d/pipe_junction.dae (Resolved: true)
Show at runtime	true
Lock	false
Public	true
Position and size	
x	0.0
y	0.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Usd ignore generated object	false
Advanced	
Show in	2D and 3D
Show name	false
Internal lights	Off

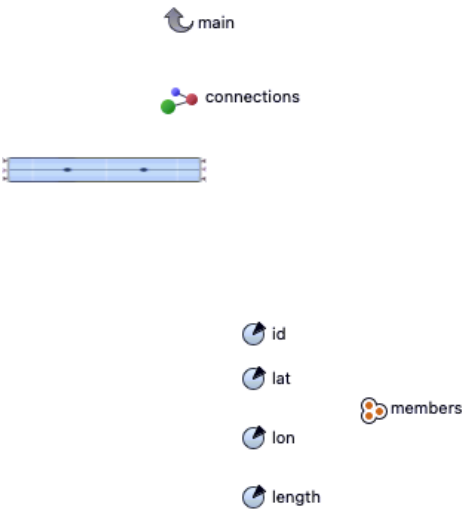
Link to agents: connections

Name	Value
General	

Name	Value
Show at runtime	true
Show name	true
Communication	
Message type	Object
Animation	
Draw line	false

Agent Type: Subway

Name	Value
Agent in flowcharts	
Use in flowcharts as	Agent
Dimensions and movement	
Speed	(10 : MPS)
Rotate animation towards movement	true
Rotate vertically (incline on ramps)	false
Space and network	
Space Type	Continuous
Layout Type	User-defined
Network type	User-defined
Enable steps	false
Advanced Java	
Generic	false
Omniverse	
Usd binding type	None
Advanced	
Flowchart block	false
Logging	true
Auto-create datasets	true
AOC_DATASETS_UPDATE_TIME_PROPERTIES	- Recurring Event Properties
Limit the number of data samples	false



Scale: scale

Name	Value
General	
Unit	meters
Scale	10.0
Type	Defined graphically
Length, pixels	100.0
Show at runtime	false
Lock	false
Public	false
Position and size	
x	0.0
y	-150.0
Rotation	0.0

Parameter: id

Name	Value
General	
Array	false
Type	int
Show at runtime	true
Show name	true
Value editor	
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Parameter: lat

Name	Value
General	
Array	false
Type	double
Show at runtime	true
Show name	true
Value editor	
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Parameter: lon

Name	Value
General	
Array	false
Type	double

Name	Value
Show at runtime	true
Show name	true
Value editor	
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Parameter: length

Name	Value
General	
Array	false
Type	double
Show at runtime	true
Show name	true
Value editor	
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Collection: members

Name	Value
General	
Initial contents	{}
Initial contents	{}
Element class	Subway
Collection class	ArrayList
Show at runtime	true
Show name	true
Advanced	
Access type	public
Save in snapshot	true

3D Object: passengerCar1

Name	Value
General	
Scale	0.5
Auto scale	true
File Name	ResourceReference: predictionmodelv1.3d/passenger_car_1.dae (Resolved: true)
Show at runtime	true
Lock	false
Public	true

Name	Value
Position and size	
x	0.0
y	0.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Usd ignore generated object	false
Advanced	
Show in	2D and 3D
Show name	false
Internal lights	Off

Link to agents: connections

Name	Value
General	
Show at runtime	true
Show name	true
Communication	
Message type	Object
Animation	
Draw line	false

Simulation Experiment: Simulation

Name	Value
General	
Bypass Initial Simulation Screen	false
Maximum available memory	16384
Agent type	Main
Model time	
Execution mode	Real time with scale
Real time scale	1.0
Stop option	Stop at specified time
Initial time	0.0
Final time	100.0
Initial date	Sat Nov 23 00:00:00 GMT 2024
Randomness	
Random Number Generation Type	Fixed seed (reproducible simulation runs)
Seed value	1
Window	
Title	PredictionModelV1 : Simulation
Enable zoom and panning	true
Enable developer panel	true
Show developer panel on start	false
Advanced	
Load root from snapshot	false

PredictionModelV1

Text: text

Name	Value
General	
Show at runtime	true
Lock	false
Text	
Text	PredictionModelV1
Appearance	
Color	royalBlue
Alignment	LEFT
Position and size	
x	50.0
y	30.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Database: Database

Name	Value
Log	
Logging	true

Database Table: seoul_subsidence

Name	Value
Advanced	
Cached database table	false

null	null	null	null	null
al_id	INTEGER	+	+	
lat	DOUBLE	-	-	
lon	DOUBLE	-	-	

Database Table: seoul_subwaystation

Name	Value
Advanced	
Cached database table	false

null	null	null	null	null
------	------	------	------	------

null	null	null	null	null
al_id	INTEGER	+	+	
lat	DOUBLE	-	-	
lon	DOUBLE	-	-	

Database Table: seoul_ntree

Name	Value
Advanced	
Cached database table	false

null	null	null	null	null
al_id	INTEGER	+	+	
lon	DOUBLE	-	-	
lat	DOUBLE	-	-	

Database Table: seoul_btree

Name	Value
Advanced	
Cached database table	false

null	null	null	null	null
al_id	INTEGER	+	+	
lon	DOUBLE	-	-	
lat	DOUBLE	-	-	

Database Table: seoul_mtree

Name	Value
Advanced	
Cached database table	false

null	null	null	null	null
al_id	INTEGER	+	+	
lon	DOUBLE	-	-	
lat	DOUBLE	-	-	

Database Table: seoul_pipe

Name	Value
Advanced	
Cached database table	false

null	null	null	null	null
------	------	------	------	------

null	null	null	null	null
al_id	INTEGER	+	+	
lon	DOUBLE	-	-	
lat	DOUBLE	-	-	
year	INTEGER	-	-	